SIMATS ENGINEERING
ASSIGNMENT

COURSE: MACHINE LEARNING (ITA 0603)

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Unit – Artificial Neural Networks: Perceptron, Multilayer Networks, Radial Basis Function (RBF) Networks, Instance-Based Learning

Develop an RBF Network to Predict Industrial Machine Failures Using Sensor Data

1. Problem Understanding

Objective:

To design and build a Radial Basis Function (RBF) Neural Network that predicts industrial machine failure from real-time sensor readings (vibration, temperature, pressure, current draw, RPM), enabling predictive maintenance instead of reactive or purely time-based maintenance.

Problem Statement:

Given historical sensor data collected from industrial machines — including vibration amplitude, operating temperature, motor current, pressure, and rotational speed — predict whether a machine is likely to fail within a defined future window (binary classification: Failure / No-Failure), or classify the machine's health state as Normal, Warning, or Critical (multi-class classification). An RBF Network is well suited to this problem because its Gaussian hidden units naturally model localized regions of the sensor-value space that correspond to specific fault signatures, making it effective at capturing non-linear boundaries between normal and faulty operating conditions.

Key Objectives:

- Identify which sensor readings are the strongest early indicators of impending machine failure (e.g., abnormal vibration spikes, temperature drift).
- Build an RBF Network that classifies machine health status with high recall on the failure class, since missing an actual failure is far costlier than a false alarm.
- Enable predictive maintenance — flagging machines for inspection before a breakdown occurs, reducing unplanned downtime.
- Present results through clear visualizations and a reproducible, well-documented workflow suitable for deployment on a factory floor dashboard.

2. Data Collection & Preparation

Dataset:

A structured sensor dataset such as the NASA CMAPSS Turbofan Engine Degradation dataset, the AI4I 2020 Predictive Maintenance dataset (UCI Machine Learning Repository), or an equivalent industrial IoT sensor log is used. Each row represents one reading (or one aggregated time window) from a machine, and columns represent the sensor features listed below.

Feature	Description	Type
Vibration (mm/s)	RMS vibration amplitude from an accelerometer	Numeric
Temperature (°C)	Operating temperature of the motor/bearing	Numeric

Pressure (bar)	Hydraulic/pneumatic system pressure	Numeric
Current Draw (A)	Motor current, reflects load and mechanical strain	Numeric
RPM	Rotational speed of the machine shaft	Numeric
Runtime Hours	Cumulative operating hours since last maintenance	Numeric
Maintenance Type	Preventive / Corrective / None	Categorical
Failure Status	Target variable to be predicted	Categorical

Preprocessing Steps:

- Handling missing sensor readings — numeric columns imputed with the rolling median (robust to sensor dropout spikes); categorical columns imputed with the mode.
- Encoding categorical variables (e.g., Maintenance Type, Failure Status) using Label Encoding for ordinal-like fields or One-Hot Encoding for nominal fields.
- Feature scaling — RBF Networks are distance-based (Gaussian activations depend directly on Euclidean distance to each center), so StandardScaler (zero mean, unit variance) is applied to every numeric sensor feature; skipping this step would let high-magnitude features like RPM dominate the distance calculation.
- Removing duplicate readings and detecting/handling outliers (e.g., physically impossible negative pressure or temperature values outside sensor range) using IQR-based capping.
- Splitting the dataset into training (80%) and testing (20%) sets using train_test_split, with stratification on the Failure Status column since failure events are typically a minority class.

Sample Preprocessing Code:

```
import pandas as pd
import numpy as np

df = pd.read_csv("machine_sensor_data.csv")
print(df.shape)
print(df.isnull().sum())

# Handle missing values
df.fillna(df.median(numeric_only=True), inplace=True)

# Encode target and categorical columns
df['maintenance_type'] = df['maintenance_type'].map({'Preventive':0,'Corrective':1,'None':2})
df['failure_status']   = df['failure_status'].map({'No':0,'Yes':1})

# Feature scaling (critical for RBF distance-based activations)
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

X = df.drop('failure_status', axis=1)
y = df['failure_status']

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42, stratify=y)
```

3. Analysis Using NumPy/Pandas

Pandas is used to explore the dataset's structure, summary statistics, and relationships between sensor readings, while NumPy supports the underlying numerical computations (correlation, distance calculations, array operations for the RBF hidden layer).

```
# Descriptive statistics
print(df.describe())

# Correlation of each sensor with the failure status
correlation = df.corr(numeric_only=True)['failure_status'].sort_values(ascending=False)
print(correlation)

# Average sensor readings grouped by failure status
print(df.groupby('failure_status').mean(numeric_only=True))

# NumPy-based summary of vibration readings
vibration = df['vibration'].to_numpy()
print("Mean:", np.mean(vibration), "Std Dev:", np.std(vibration))
```

Expected Insight:

This analysis typically reveals that Vibration and Temperature have the strongest positive correlation with Failure Status, since bearing wear and overheating are leading physical precursors to mechanical breakdown, followed by Current Draw (which rises as a motor works harder against increasing friction). Runtime Hours also shows a positive trend, reflecting natural wear-and-tear — together these guide which sensors are most important to retain and to place near RBF cluster centers representing 'faulty' regions of the input space.

4. Model Selection & Training

Since the target (Failure / No-Failure) is best modeled as a non-linear boundary in sensor space, a Radial Basis Function (RBF) Network is selected. An RBF Network has three layers: an input layer (the scaled sensor features), a hidden layer of Gaussian radial basis functions whose centers are typically found via K-Means clustering on the training data and whose spread (σ) controls how localized each unit's response is, and a linear output layer whose weights are learned to combine the hidden-layer activations into a final prediction. A Support Vector Machine with an RBF kernel is used as a comparison baseline, since it shares the same Gaussian-similarity intuition.

Sample Implementation Code:

```
from sklearn.cluster import KMeans
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score
import numpy as np

# --- RBF hidden layer ---
n_centers = 15
kmeans = KMeans(n_clusters=n_centers, random_state=42, n_init=10)
kmeans.fit(X_train)
centers = kmeans.cluster_centers_
```

```
# Spread (sigma) — average distance between centers, a common heuristic
from scipy.spatial.distance import cdist
d_max = cdist(centers, centers).max()
sigma = d_max / np.sqrt(2 * n_centers)

def rbf_activation(X, centers, sigma):
    dist = cdist(X, centers)
    return np.exp(-(dist ** 2) / (2 * sigma ** 2))

Phi_train = rbf_activation(X_train, centers, sigma)
Phi_test  = rbf_activation(X_test, centers, sigma)

# --- Linear output layer trained on RBF activations ---
output_layer = LogisticRegression(max_iter=1000)
output_layer.fit(Phi_train, y_train)
rbf_pred = output_layer.predict(Phi_test)
rbf_proba = output_layer.predict_proba(Phi_test)[:, 1]

# --- Baseline comparison: SVM with RBF kernel ---
svm = SVC(kernel='rbf', probability=True, random_state=42)
svm.fit(X_train, y_train)
svm_pred = svm.predict(X_test)

print("RBF Network Accuracy:", accuracy_score(y_test, rbf_pred))
print("RBF Network Recall (failure class):", recall_score(y_test, rbf_pred))
print("RBF Network F1-Score:", f1_score(y_test, rbf_pred))
print("RBF Network ROC-AUC:", roc_auc_score(y_test, rbf_proba))
```

Model Evaluation:

Metric	Purpose
Accuracy	Overall proportion of correctly classified machine states
Precision	Of machines flagged as 'will fail', how many genuinely failed — controls false-alarm rate
Recall (Sensitivity)	Of machines that actually failed, how many the model caught — the priority metric here, since a missed failure risks costly downtime or safety hazards
F1-Score	Harmonic balance between precision and recall
ROC-AUC	Overall ability to separate 'will fail' from 'normal' across all decision thresholds

The RBF Network is expected to outperform a plain linear classifier because its Gaussian hidden units can represent tight, localized 'fault signatures' in sensor space (e.g., high vibration combined with high temperature) that a linear boundary cannot separate cleanly, while remaining faster to train than a deep neural network and easier to interpret through its cluster centers.

5. Visualization & Interpretation

```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix, RocCurveDisplay
```

```
# Correlation heatmap of sensor features
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap of Machine Sensor Features")
plt.show()

# Vibration vs Temperature colored by failure status
sns.scatterplot(x='vibration', y='temperature', hue='failure_status', data=df, palette='Set1')
plt.title("Vibration vs Temperature — Failure Regions")
plt.show()

# Confusion matrix for the RBF Network
cm = confusion_matrix(y_test, rbf_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix — RBF Network")
plt.show()

# ROC curve
RocCurveDisplay.from_predictions(y_test, rbf_proba)
plt.title("ROC Curve — RBF Network")
plt.show()
```

Interpretation of Visuals:

- Correlation Heatmap – visually confirms that Vibration and Temperature have the darkest (strongest) correlation with Failure Status, guiding sensor prioritization for maintenance teams.
- Vibration-vs-Temperature Scatter Plot – reveals a distinct cluster of failure points at high vibration and high temperature, exactly the kind of localized region an RBF hidden unit is designed to capture with its Gaussian center.
- Confusion Matrix – shows how many actual failures were correctly caught (true positives) versus missed (false negatives); minimizing false negatives is the priority for a predictive-maintenance system.
- ROC Curve – summarizes the model's ability to separate failure from normal operation across all thresholds, letting engineers choose an operating threshold that favors recall (catching more failures) even at some cost to precision.

6. Innovation & SDG Mapping

This project directly aligns with United Nations Sustainable Development Goal 9 (SDG 9: Industry, Innovation and Infrastructure), which aims to 'build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation'.

Real-World Relevance:

- Predictive Maintenance – the RBF Network flags machines likely to fail before a breakdown occurs, shifting industry from costly reactive repairs and rigid time-based maintenance schedules toward efficient, condition-based maintenance.
- Resilient Infrastructure – reducing unplanned downtime keeps production lines and critical industrial infrastructure (power, water treatment, manufacturing) running reliably, directly supporting SDG 9's infrastructure-resilience target.
- Resource-Efficient Industrialization – early fault detection prevents cascading mechanical damage, reducing material waste, energy loss, and spare-part consumption — supporting sustainable industrial practices.
- Fostering Innovation – demonstrates how a classical, computationally light neural architecture (RBF) can be deployed on edge devices or embedded industrial IoT controllers where a full deep-

learning model would be too resource-intensive, making advanced predictive analytics accessible to smaller manufacturers.

Innovation Aspect:

Rather than treating machine failure prediction as a black box, mapping the RBF Network's cluster centers back to physical sensor ranges turns the model into an interpretable diagnostic tool — maintenance engineers can see exactly which combination of vibration, temperature, and load values a given 'fault center' represents, moving beyond a simple failure/no-failure flag toward genuine root-cause-aware predictive maintenance.

7. Documentation & Conclusion

Summary of Workflow:

Sensor Data Collection → Preprocessing (cleaning, encoding, scaling) → Exploratory Analysis (Pandas/NumPy) → Visualization (Matplotlib/Seaborn) → RBF Network Training (K-Means centers + Gaussian activations + linear output layer, compared against an SVM-RBF baseline) → Evaluation (Accuracy, Precision, Recall, F1, ROC-AUC) → Insight Generation & SDG Mapping.

Conclusion:

The designed RBF Network demonstrates that industrial machine failure can be reasonably predicted from measurable sensor indicators such as vibration, temperature, and current draw, with the Gaussian hidden layer able to capture the localized, non-linear sensor combinations that precede mechanical failure more effectively than a purely linear model. Beyond raw predictive accuracy, the project's real value lies in enabling early, targeted maintenance intervention — reducing unplanned downtime, conserving resources, and supporting more resilient, innovative industrial infrastructure, directly advancing SDG 9: Industry, Innovation and Infrastructure.

Future Scope:

- Incorporate additional sensor streams (acoustic emission, oil-particle analysis) to further improve early-warning accuracy.
- Deploy the trained RBF Network on an edge/IoT controller for real-time, on-machine failure scoring rather than periodic batch analysis.
- Experiment with a hybrid RBF-LSTM architecture to capture temporal degradation trends in addition to instantaneous sensor snapshots.